

Project Design Phase

Solution Architecture

Project: Freelancer Web Application (MERN Stack)

Date	10 April 2026
Team ID	Your Team ID
Project Name	Freelancer Web Application
Maximum Marks	4 Marks

Solution Architecture (Introduction)

Solution architecture bridges the gap between business problems and technology solutions. It defines how different components of the system interact and ensures scalability, performance, and security. For the Freelancer Web Application, the architecture is designed using the MERN stack to provide a full-stack solution.

The main goals of this architecture are:

- To provide a scalable and efficient platform for freelancers and clients
- To ensure secure communication and transactions
- To define system structure and data flow clearly
- To support future enhancements and integrations

System Architecture Overview

The system follows a 3-tier architecture consisting of:

- Presentation Layer (Frontend - React.js)
- Application Layer (Backend - Node.js & Express.js)
- Data Layer (Database - MongoDB)

Users interact with the frontend, which communicates with the backend APIs. The backend processes requests and interacts with the database to store and retrieve data.

Key Components

Component	Description
Frontend	React-based UI for freelancers and clients
Backend	Node.js & Express.js for APIs and logic
Database	MongoDB for storing users, jobs, messages
Authentication	JWT-based secure login system
Messaging	Socket.io for real-time chat
Payment	Integration with Razorpay/Stripe
Storage	Cloudinary for file uploads

Data Flow

1. User logs in/registers via frontend
2. Request sent to backend API
3. Backend validates and interacts with database
4. Data returned to frontend
5. Real-time updates handled via Socket.io

Core Features Supported

- User authentication and profile management
- Job posting and bidding system
- Real-time messaging
- Secure payment processing
- Review and rating system

Security Considerations

- JWT authentication for secure sessions
- Password encryption using bcrypt
- HTTPS secure communication
- Input validation and error handling

Deployment & Infrastructure

- Frontend hosted on Vercel
- Backend hosted on Render/Node server
- Database hosted on MongoDB Atlas
- Cloud storage using Cloudinary

Conclusion

The solution architecture ensures a robust, scalable, and secure Freelancer Web Application. By leveraging the MERN stack, the system provides seamless interaction between frontend, backend, and database, delivering an efficient platform for freelancers and clients.